

Seasonal Agriculture Performance Analysis

VOIS AICTE Major Project | Data Analytics

Introduction

This project analyzes agricultural activities across seasons and investigates patterns in productivity, economics, resource usage, environmental conditions and disease/pest risk.

Objectives

Explore, clean and analyze the dataset; compare seasons; investigate relationships; apply visualization and statistical techniques; and produce evidence-based recommendations.

Dataset

The dataset contains 4,000 rows and 28 columns, covering 3 seasons, 8 crops and 8 states. There are 120 missing cells before cleaning and no duplicate rows.

Technology

Python, Pandas, NumPy, Matplotlib, SciPy, Jupyter Notebook and Streamlit.

Methodology

Data loading → data quality assessment → median imputation for selected numeric missing values → descriptive statistics → seasonal KPI comparison → crop-season analysis → correlation → ANOVA → recommendations.

Seasonal Results

Season	Yield t/ha	Revenue	Cost	Profit	Water m³	Efficiency	Risk %
Kharif	5.64	■710,719	■531,804	■178,915	6,102	5.89	54.47
Rabi	5.08	■601,526	■513,837	■87,689	5,847	5.19	40.48
Zaid	4.67	■519,172	■543,977	■-24,805	6,420	4.41	38.22

Key Findings

Kharif leads average yield at 5.64 t/ha and average profit at ■178,915. Zaid shows negative average profit of ■-24,805; its water use is highest and water efficiency lowest among the three seasons. Kharif also has the highest average disease/pest risk.

Recommendations

Review Zaid cost structure; compare irrigation efficiency; prioritize resilient crop-season combinations; strengthen pest monitoring; analyze district-level exceptions; and maintain a repeatable dashboard for monitoring.

Future Scope

Predictive yield modeling, weather/market integration, crop recommendation, forecasting, automated BI dashboards and disease/pest early warning.